

PROJECT DESIGN PHASE

Solution Architecture

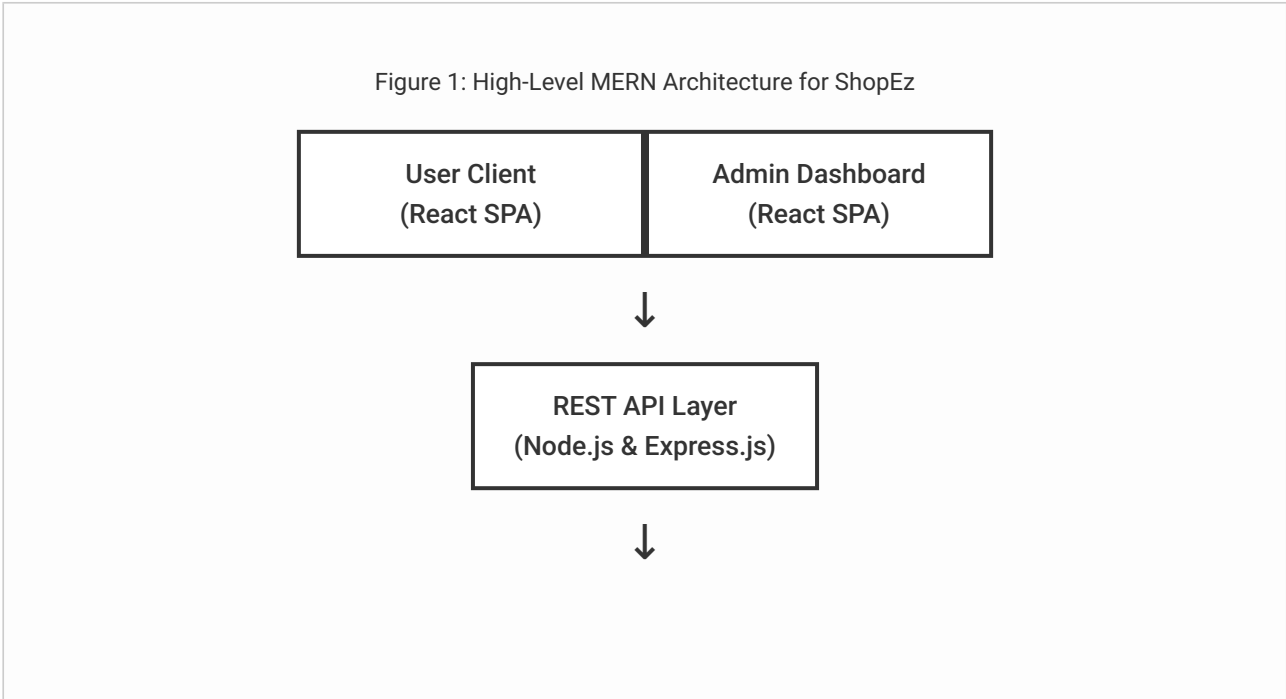
Date	18 June 2026
Team ID	
Project Name	ShopEz (Full Stack MERN-Based E-Commerce)
Maximum Marks	4 Marks

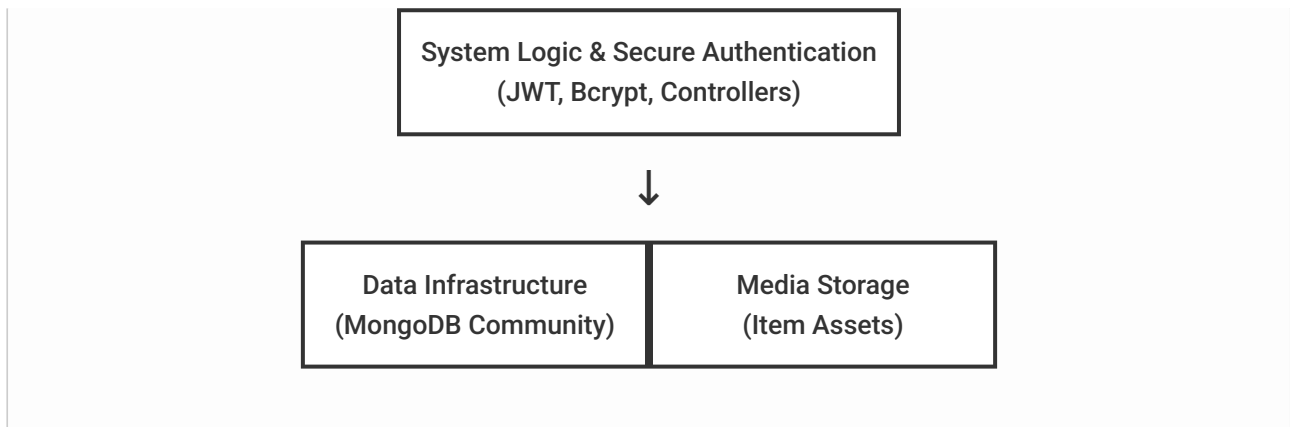
Solution Architecture:

High-quality solution architecture serves as the blueprint for connecting technical implementation with core business goals. For the ShopEz platform, the framework is engineered to facilitate massive product catalogs and high-security transactions leveraging the full MERN stack capabilities.

- Utilize the MERN ecosystem to address complex e-commerce browsing and administrative needs.
- Implement a robust database schema for managing Users, Inventory, and Transactions via MongoDB.
- Guarantee encrypted API interaction through Express.js and industry-standard JWT protocols.
- Ensure a fluid user experience with a high-performance React Single Page Application (SPA).

ShopEz - Solution Architecture Diagram:





Component Descriptions:

Frontend Layer (React): Decoupled interfaces tailored for varying user roles; providing a seamless shopping journey for guests and specialized management tools for owners via reusable UI components.

Server & API Layer (Node/Express): Manages authenticated HTTP communication, data validation, and route orchestration. It ensures persistent availability and rapid data delivery through non-blocking asynchronous operations.

Storage & Data Layer (MongoDB): Employs flexible NoSQL schemas to handle dynamic catalogs and order details, complemented by file storage for high-quality product images.

Operational Data Flow: Frontend actions trigger RESTful requests; the backend authenticates user sessions via secure tokens, interacts with the MongoDB cluster, and delivers JSON data for real-time React rendering.